

1.4 Arc Length Worksheet

1. Find the exact arc length of the curve over the given interval: $y = x^2$ from $x = 1$ to $x = 3$.

(a) Set up, but do not evaluate, an integral with respect to x .

We note that $y' = 2x$ which is continuous from $x = 1$ to $x = 3$.

Therefore the arc length is given by:
$$L = \int_1^3 \sqrt{1 + (2x)^2} dx = \int_1^3 \sqrt{1 + 4x^2} dx$$

We do not yet have the ability to evaluate such an integral. More advance integration techniques in Ch. 2 will allow us to do so.

(b) Set up, but do not evaluate, an integral with respect to y .

We must solve the equation for x . $y = x^2 \rightarrow x = \sqrt{y}$.

We note that $y(1) = (1)^2 = 1$ & $y(3) = (3)^2 = 9$, therefore we are interested in the arc length of $x = \sqrt{y}$ from $y = 1$ to $y = 9$

We also note that $\frac{dx}{dy} = \frac{1}{2}(y)^{-\frac{1}{2}} = \frac{1}{2\sqrt{y}}$ is continuous from $y = 1$ to $y = 9$.

Therefore the arc length is given by:
$$L = \int_1^9 \sqrt{1 + \left(\frac{1}{2\sqrt{y}}\right)^2} dy = \int_1^9 \sqrt{1 + \frac{1}{4y}} dy$$

2. Determine the length of $y = \ln(\sec(x))$ between $0 \leq x \leq \frac{\pi}{4}$.

(Hint: Although we have not shown this in class yet it is true that $\int \sec(x) dx = \ln|\sec(x) + \tan(x)| + C$)

We see that $y' = \frac{\sec(x)\tan(x)}{\sec(x)} = \tan(x)$ which is continuous for $0 \leq x \leq \frac{\pi}{4}$.

Therefore the arc length of the curve is given by:
$$L = \int_0^{\pi/4} \sqrt{1 + \tan^2(x)} dx$$

We then have $L = \int_0^{\pi/4} \sqrt{1 + \tan^2(x)} dx = \int_0^{\pi/4} \sqrt{\sec^2(x)} dx = \int_0^{\pi/4} |\sec(x)| dx$. Since $0 \leq x \leq \frac{\pi}{4}$ we know that $\sec(x) \geq 0$ and therefore $|\sec(x)| = \sec(x)$. So we have:

$$\begin{aligned} L &= \int_0^{\pi/4} \sqrt{1 + \tan^2(x)} dx = \int_0^{\pi/4} \sqrt{\sec^2(x)} dx = \int_0^{\pi/4} |\sec(x)| dx = \int_0^{\pi/4} \sec(x) dx = \ln|\sec(x) + \tan(x)| \Big|_0^{\pi/4} \\ &= \ln|\sqrt{2} + 1| - \ln|1 + 0| = \ln(\sqrt{2} + 1) \end{aligned}$$

3. Setup, but do not evaluate, an integral which represents the length of the curve $y = \sqrt{x}$ on $[0, 4]$.

We note that $y' = \frac{1}{2\sqrt{x}}$ is NOT continuous on $[0, 4]$ and therefore we cannot set up this integral with respect to x . We instead will set the integral up with respect to y .

$$y = \sqrt{x} \rightarrow x = y^2 \text{ on } [0, 2].$$

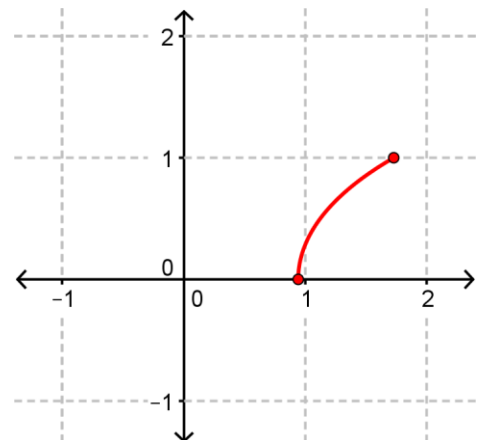
We note that $x' = 2y$ is continuous on $[0, 2]$.

Therefore, the arc length is given by $L = \int_0^2 \sqrt{1 + (2y)^2} dy = \int_0^2 \sqrt{1 + 4y^2} dy$.

4. Find the exact arc length of the curve over the given interval. $x = \frac{1}{3}(y^2 + 2)^{3/2}$ from $y = 0$ to $y = 1$.

$$\frac{dx}{dy} = \frac{1}{3} \cdot \frac{3}{2} (y^2 + 2)^{1/2} (2y) = y\sqrt{y^2 + 2} \text{ is continuous on } [0, 1].$$

$$\begin{aligned} L &= \int_0^1 \sqrt{1 + \left[y\sqrt{y^2 + 2}\right]^2} dy = \int_0^1 \sqrt{1 + y^2(y^2 + 2)} dy = \int_0^1 \sqrt{1 + y^4 + 2y^2} dy \\ &= \int_0^1 \sqrt{(y^2 + 1)^2} dy = \int_0^1 (y^2 + 1) dy = \left(\frac{y^3}{3} + y\right) \Big|_0^1 = \frac{1}{3} + 1 = \frac{4}{3} \end{aligned}$$

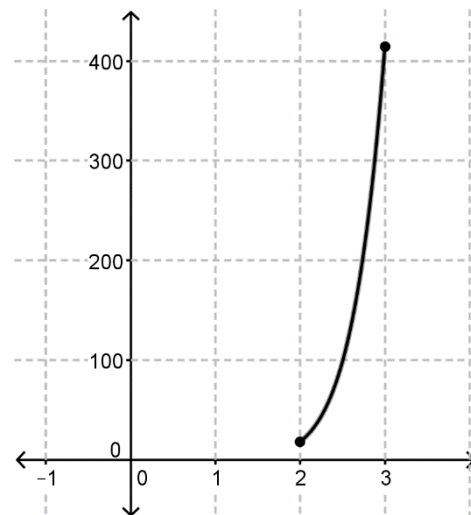


5. Find the exact arc length of the curve over the given interval. $y = \frac{x^6 + 8}{16x^2}$ from $x = 2$ to $x = 3$.

We note that $y = \frac{1}{16}x^4 + \frac{1}{2}x^{-2}$.

So $y' = \frac{1}{4}x^3 - x^{-3} = \frac{x^3}{4} - \frac{1}{x^3}$ is continuous on $[2, 3]$.

$$\begin{aligned} L &= \int_2^3 \sqrt{1 + \left(\frac{x^3}{4} - \frac{1}{x^3}\right)^2} dx = \int_2^3 \sqrt{1 + \frac{x^6}{16} + 2\left(\frac{x^3}{4}\right)\left(-\frac{1}{x^3}\right) + \frac{1}{x^6}} dx \\ &= \int_2^3 \sqrt{\frac{x^6}{16} + \frac{1}{2} + \frac{1}{x^6}} dx = \int_2^3 \sqrt{\left(\frac{x^3}{4} + \frac{1}{x^3}\right)^2} dx = \int_2^3 \left(\frac{x^3}{4} + \frac{1}{x^3}\right) dx \\ &= \left(\frac{x^4}{16} - \frac{x^{-2}}{2}\right) \Big|_2^3 = \left(\frac{81}{16} - \frac{1}{18}\right) - \left(1 - \frac{1}{8}\right) = \frac{595}{144} \end{aligned}$$



6. Kyle is setting up an integral with respect to y which is supposed to find the arc length of the curve $y = x^3 + 4$ on $[-1, 1]$. He sets up the following integral with respect to y

$$L = \int_3^5 \sqrt{1 + \left(\frac{1}{3}(y-4)^{-2/3}\right)^2} dy$$

Kyle's professor tells him that his integral isn't guaranteed to represent the arc length. Why is this?

The issue here is $x' = \frac{1}{3}(y-4)^{-2/3}$ is not continuous at on $[-4, 5]$, specifically at $y = 4$.

Since the conditions for the arc length formula are not met, we cannot guarantee that this integral represents the arc length of $y = x^3 + 4$ on $[-1, 1]$.

7. Find the exact arc length of $24xy = y^4 + 48$ from $y = 2$ to $y = 4$.

We note that $24xy = y^4 + 48 \rightarrow x = \frac{y^4 + 48}{24y} = \frac{1}{24}y^3 + \frac{2}{y}$. So, $y' = \frac{1}{8}y^2 - \frac{2}{y^2}$ is continuous on $[2, 4]$.

Therefore,

$$\begin{aligned} L &= \int_2^4 \sqrt{1 + \left(\frac{1}{8}y^2 - \frac{2}{y^2}\right)^2} dy = \int_2^4 \sqrt{1 + \frac{1}{64}y^4 - 2\left(\frac{1}{8}y^2 \cdot \frac{2}{y^2}\right) + \frac{4}{y^4}} dy = \int_2^4 \sqrt{\frac{1}{64}y^4 + \frac{1}{2} + \frac{4}{y^4}} dy \\ &= \int_2^4 \sqrt{\left(\frac{1}{8}y^2 + \frac{2}{y^2}\right)^2} dy = \int_2^4 \left|\frac{1}{8}y^2 + \frac{2}{y^2}\right| dy \end{aligned}$$

We can see that $\frac{1}{8}y^2 + \frac{2}{y^2}$ is positive for all values on $[2, 4]$, so we can drop the absolute value bars.

$$\begin{aligned} \int_2^4 \left|\frac{1}{8}y^2 + \frac{2}{y^2}\right| dy &= \int_2^4 \left(\frac{1}{8}y^2 + \frac{2}{y^2}\right) dy = \left(\frac{1}{24}y^3 - \frac{2}{y}\right) \Big|_2^4 = \left(\frac{1}{24}(64) - \frac{1}{2}\right) - \left(\frac{1}{24}(8) - 1\right) \\ &= \left(\frac{64}{24} - \frac{12}{24}\right) - \left(\frac{8}{24} - \frac{24}{24}\right) = \frac{68}{24} = \frac{34}{12} = \frac{17}{6} \end{aligned}$$

8. Find the exact arc length of $y = 3x^{3/2} - 1$ from $x = 0$ to $x = 1$.

We note that $y' = \frac{9}{4}x^{1/2}$ is continuous on $[0, 1]$.

Therefore,

$$L = \int_0^1 \sqrt{1 + \left(\frac{9}{4}x^{1/2}\right)^2} dx = \int_0^1 \sqrt{1 + \frac{81}{16}x} dx$$

We proceed with a u -sub. Let $u = 1 + \frac{81}{16}x$ and this implies $du = \frac{81}{16}dx \rightarrow \frac{16}{81}du = dx$

Our bounds become $u(0) = 1$ & $u(1) = \frac{97}{16}$.

$$\int_0^1 \sqrt{1 + \frac{81}{16}x} dx = \int_1^{97/16} \frac{16}{81} \sqrt{u} du = \frac{16}{81} \left(\frac{2}{3}\right) u^{3/2} \Big|_1^{97/16} = \frac{32}{243} \left(\frac{97}{16}\right)^{3/2} - \frac{32}{243}$$